

When It's Too Hot, Animals Get Confused, Cranky, and Forgetful

From: Smithsonian Science-Nature Mined: 2026-06-09 3 min read

Have you ever felt so hot that you couldn't think straight? It turns out animals feel the same way and it can cause big problems. Scientists have found that during **heat waves**, many creatures get confused, aggressive, or just plain forgetful. In South Africa, little birds called **pieb babbler** were given a puzzle: a tasty worm behind a clear wall. On cool days, they quickly figured out to go around it. But when the temperature soared, they just kept pecking at the plastic, unable to learn. Other birds, like **zebra finches**, also banged their beaks uselessly against tubes when it was hot. Even fish struggle. Male **guppies** in warm water couldn't find their way through a maze, even with a female guppy waiting at the end. **Bumblebees** forgot which flowers gave sweet nectar when temperatures rose. And dogs? A study of nearly 70,000 dog bites in U.S. cities found bites were 10% more likely on 90-degree days than on 60-degree days. It's not just pets. Wild chamois, a kind of goat-antelope, get into more fights over food when it's hot. One study predicts their aggression could jump 50% by 2080 because of **climate change**. Why does heat scramble brains? For animals that can't control their body temperature, like fish and insects, a hotter brain may slow down nerves, hurting memory and learning. For birds and mammals, staying cool takes so much energy they have less brainpower left for thinking. This isn't just bad for the animals. If bees forget which crops to pollinate, our food supply could suffer. If birds can't find food for their chicks, fewer survive. As **heat waves** become more common, a sharp mind might be the difference between life and death for them, and maybe for us too.

Word Treasure

heat waves -- Long stretches of unusually hot weather that can be dangerous for living things.

pieb babbler -- Social birds from South Africa that live in groups and solve simple problems to find food.

zebra finches -- Small Australian songbirds with striped patterns on their heads and chests.

guppies -- Tiny, brightly colored fish often found in warm, tropical waters.

bumblebees -- Fat, fuzzy bees that carry pollen from flower to flower, helping plants make fruit.

climate change -- A big shift in the Earth's weather happening over many years, making the planet warmer.

Background you need

Pied babblers are tiny teamwork birds that live in South Africa's dry savannas. Chamois are sure-footed goat-antelopes that climb rocky mountains in Europe.

Many animals cannot sweat or pant well enough to keep their brains cool. When temperatures rise too high, their thinking gets cloudy -- just like a student trying to study in a hot, stuffy room.

Break it down

WHO: Birds (pied babblers, zebra finches), fish (guppies), insects (bumblebees), dogs, and chamois -- plus the scientists studying them.

WHAT: During heat waves, these animals get confused, forgetful, and start more fights.

WHERE: In South Africa, U.S. cities, and mountain habitats where the animals live.

WHY: In cold-blooded creatures, a hot brain slows nerves; in warm-blooded ones, keeping the body cool steals energy needed for thinking.

Why it matters

If bees forget which flowers to pollinate, the fruits and vegetables that end up on your dinner plate could become much harder to grow.

Test what you remember

1. Which birds kept pecking uselessly at a clear wall on hot days?
 - (A) Pied babblers
 - (B) Zebra finches
 - (C) Pigeons
 - (D) Eagles
2. What kind of fish couldn't solve a maze when the water was warm?
 - (A) Male guppies
 - (B) Goldfish
 - (C) Sharks
 - (D) Salmon
3. On very hot days, dog bites become more likely by about how much compared to cool days?
 - (A) 10% more likely
 - (B) 50% more likely
 - (C) 5% less likely
 - (D) The same
4. The article describes a goat-antelope that fights more when it's hot. What is it called?
 - (A) Chamois
 - (B) Springbok
 - (C) Ibex
 - (D) Impala
5. Why does heat make it harder for insects like bumblebees to remember things?
 - (A) Their brains get too hot and nerves slow down
 - (B) They run out of water
 - (C) They hibernate
 - (D) They lose their wings
6. What could increase by 50% among chamois by the year 2080, according to one study?
 - (A) Aggression
 - (B) Running speed
 - (C) Body size
 - (D) Sleeping time

Answer key (try the quiz first!): 1: A 2: A 3: A 4: A 5: A 6: A

Questions to think about

1. **Animal lovers worry:** They see heat making pets grumpy and wild creatures starving, and they want more cooling shelters and less pollution to protect them.
2. **Scientists treat it as a warning:** They believe studying how heat scrambles animal brains can help us protect food supplies and plan for a warmer future.

MY THOUGHTS (at least 20 words):